

Ricardo (Zhicong) Xie

Manufacturing / Site Project Intern | CAD-to-Part, Workshop Practice and Coordination

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LinkedIn: <https://www.linkedin.com/in/ricardo-xie-6394643b4> | Portfolio: <https://zxie588-alt.github.io> | GitHub: <https://github.com/zxie588-alt>

NZ student visa | Full-time eligible over scheduled summer break 17 Nov 2026 - late Feb 2027 | No employer visa support needed for this internship | Open NZ-wide

PROFILE

- Mechanical master's student with laser cutting, CNC exposure, 3D printing workflow and CAD-to-part evidence.
- Interested in manufacturing, production support and site/project engineering roles.
- Project screenshots and source links: <https://zxie588-alt.github.io>

EDUCATION

University of Auckland - Master of Engineering Studies, Mechanical Engineering | 2026 - Present

Study focus and portfolio work in mechanical, building-services, manufacturing and analysis contexts.

Study record/focus: Completed: Engineering Methods, Composite Materials, Computational Fluid Dynamics. Registered S2: Manufacturing and Industrial Systems, Smart Manufacturing.

Huazhong Agricultural University - Bachelor of Engineering, Mechanical Engineering / Mechatronics | 2021 - 2025

Mechanical design, CAD, electrotechnics, sensors, embedded control and mechatronic system integration.

PRACTICAL AND TEAM EXPERIENCE

- Engineering team leadership: led task scoping, workload split and final technical presentation in team design work.
- Workshop practice: prepared AutoCAD/DXF laser-cut geometry and checked cut parts against drawings.
- Hands-on exposure: CNC machining, workholding, cutting fluid, chip formation and 3D printing workflow.
- Lab practice: interpreted force-displacement tests and controlled a robotic arm in training sessions.
- Technical communication: former student news lead; strong project records, photography and presentation visuals.

PROJECT WORK

Manufacturing Workshop Training & DXF Preparation

Laser cutting, AutoCAD/DXF, CNC machining exposure, 3D printing workflow

- Prepared AutoCAD/DXF geometry for laser cutting and checked the cut plate against drawing intent.
- Practised CNC workflow: workholding, tool motion, cutting fluid, chip formation and shop safety.
- Reviewed machined gears and shafts for tolerance, edge quality and tool-access constraints.

Engineering Practice and Automotive Structure Training

Mechanics testing, robotic arm control, automotive structure walkthrough

- Worked with mechanics testing equipment and interpreted force-displacement results from a physical test setup.
- Practised robotic arm control and basic automation interaction in a lab setting.
- Led a small automotive-structure walkthrough covering seat/interior systems and vehicle context.

Ruggedized Outdoor IoT Sensor Module Design Package

SolidWorks, enclosure design, DFM notes, Keil C, testing workflow

- Designed a SolidWorks enclosure with gasket stack-up, screw bosses, PCB envelope and service access.
- Prepared DFM notes for CNC review, 3D printing and assembly tolerance checks.
- Reviewed load paths around corner, strap-lug and screw-boss regions; built Keil C module structure.

Dynamic Livestock Weighing System - Software Copyright Registered

STM32, HX711, embedded C, Bluetooth/serial communication

- Developed STM32/HX711 weighing firmware with real-time data transfer to a PC interface.
- Received Computer Software Copyright Registration No. 2024SR0141888 for the system.
- Supported dynamic weighing logic for improved signal stability under field-like conditions.

TECHNICAL SKILLS

- CAD-to-part: AutoCAD/DXF laser-cut prep, SolidWorks assemblies, STEP/STL export and checks.
- Workshop: CNC workflow, workholding, cutting fluid, chip formation and 3D printing workflow.
- Site support: technician communication, task tracking, workshop safety awareness and verification.